ACHARYA INSTITUTE OF TECHNOLOGY

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Computer Science & Engineering (Data Science)

PROJECT REPORT

ON

From BI to Big Data: Explain, Design & Defend

Subject Name: Big Data Analytics

Subject Code: BAD601

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1.About the Project

Introduction

In today's digital era, social media platforms generate massive amounts of data every second. X (Twitter) is one such platform where millions of users post tweets, like content, retweet, comment, and follow accounts continuously.

Managing such high-volume, high-velocity, and high-variety data requires Big Data technologies. Traditional Business Intelligence (BI) systems are not sufficient to handle real-time streaming data and unstructured content like tweets, images, and videos.

This project explains the evolution from Traditional BI systems to Big Data architecture using X (Twitter) as a case study.

Objective of the Project

The objective of this project is:

- To understand the difference between Traditional BI and Big Data.
- To analyse the 5V's of Big Data .
- To design both Traditional Data Warehouse and Hadoop-based architectures .
- To understand four types of analytics and tools used.
- To explain Big Data in a simplified way.

Need for Big Data

Traditional BI tools like Excel and SQL databases:

- Handle only structured data
- Support batch processing.
- Have limited scalability.
- Cannot process real-time streaming data.

Since Twitter generates:

- Millions of tweets per second
- Real-time trending hashtags
- Multimedia content
- User behavior logs

Big Data technologies are necessary.

2.TOOLS AND TECHNOLOGIES

2.1 Hadoop

Hadoop is an open-source framework used for distributed storage and processing of large datasets. It works on a cluster of machines and provides scalability and fault tolerance.

Components:

- HDFS (Storage Layer)
- MapReduce (Processing Layer)

Advantages:

- Handles large data
- Cost effective

- Fault tolerant
- Scalable

2.2 Apache Spark

Apache Spark is used for real-time analytics and machine learning..

Features:

- In-memory processing
- Faster than MapReduce

Used in Twitter for:

- Trend detection
- Recommendation systems
- Spam detection

Spark is used for recommendation systems and trend detection.

2.3 NoSQL Databases

NoSQL databases such as MongoDB and Cassandra store flexible data formats like JSON.

Why needed:

- Twitter data is semi-structured (JSON format
- Schema changes frequently and High scalability required.

2.4 Business Intelligence Tools

BI tools such as Power BI and Tableau are used for:

- Dashboard creation
- Report generation
- Data visualization

3. My Contribution

In this project titled "**From BI to Big Data – Twitter Case Study**", I contributed to the conceptual understanding, system design, and analytical explanation of how Big Data technologies are applied in modern social media platforms.

Firstly, I selected Twitter (X) as the case study and analysed the different types of data it generates, including structured data (user details, timestamps), semi-structured data (JSON logs), and unstructured data (tweets, images, videos). This helped in understanding the complexity and diversity of large-scale digital data.

Secondly, I performed a comparative analysis between Traditional Business Intelligence systems and Big Data systems. I identified key limitations of traditional BI such as batch processing, limited scalability, and inability to handle unstructured and real-time streaming data. I then explained the need for Big Data technologies like Hadoop, Apache Spark, and NoSQL databases to manage the 5V's of Big Data — Volume, Velocity, Variety, Veracity, and Value.

As part of the architecture task, I designed and explained two system models:

1. Traditional Data Warehouse Architecture
2. Hadoop-Based Big Data Architecture

Additionally, I analysed the four types of analytics — Descriptive, Diagnostic, Predictive, and Prescriptive — and mapped them to relevant business questions related to Twitter. I also created infographic posters and a simplified explanation of Big Data for better conceptual clarity.

Overall, my contribution includes research, comparative evaluation, architecture design, analytical mapping, and structured documentation of the project, demonstrating a clear understanding of the evolution from Traditional BI to scalable Big Data systems.

4. Code Explanation

Since this project is primarily conceptual, no actual coding implementation was performed. However, the working of a real-world Twitter Big Data system can be explained through a theoretical Big Data processing workflow.

In a real-time Twitter environment, user activities such as tweets, likes, retweets, replies, hashtag usage, and media uploads are continuously generated. This high-velocity data is first captured through data ingestion systems or streaming platforms. In practical scenarios, streaming tools like Apache Kafka collect real-time data from multiple sources.

The collected data is then stored in a distributed storage system such as HDFS (Hadoop Distributed File System). HDFS allows large volumes of structured, semi-structured, and unstructured data to be stored across multiple nodes, ensuring scalability and fault tolerance.

After storage, a processing engine like Apache Spark analyses the data in parallel. Spark performs operations such as:

* Counting total tweets and active users

* Identifying trending hashtags

* Analysing user engagement patterns

* Detecting spam or fake accounts

Machine learning models can then be applied using Spark ML lib to predict whether a tweet may go viral, recommend relevant content to users, or identify abnormal behaviour.

The processed data is stored in a NoSQL database such as MongoDB or Cassandra for fast retrieval. Finally, Business Intelligence tools like Tableau or Power BI generate dashboards that help management monitor platform performance and user trends.

This theoretical workflow demonstrates how Big Data technologies enable Twitter to handle massive real-time data, perform predictive analytics, and support intelligent decision-making at scale.

5. Architecture Overview

This project presents two different architectures to demonstrate the evolution from Traditional Business Intelligence systems to modern Big Data systems in the context of Twitter (X)

1. Traditional Data Warehouse Architecture

In the Traditional BI architecture, data from multiple operational sources such as user accounts, tweet records, and advertisement databases is collected periodically. The data undergoes an ETL (Extract, Transform, Load) process, where it is cleaned, structured, and stored in a centralized relational database.

Once stored in the data warehouse, Business Intelligence tools such as Power BI or Tableau generate reports and dashboards. This system mainly supports structured data and batch processing.

However, this architecture has several limitations when applied to Twitter:

- It cannot efficiently handle real-time tweet streams.
- It struggles with unstructured data such as text, images, and videos.
- Scaling requires expensive hardware upgrades (vertical scaling).
- Reports are generated after scheduled processing, not instantly.

Thus, Traditional BI systems are suitable for historical reporting but not for high-velocity, large-scale social media platforms

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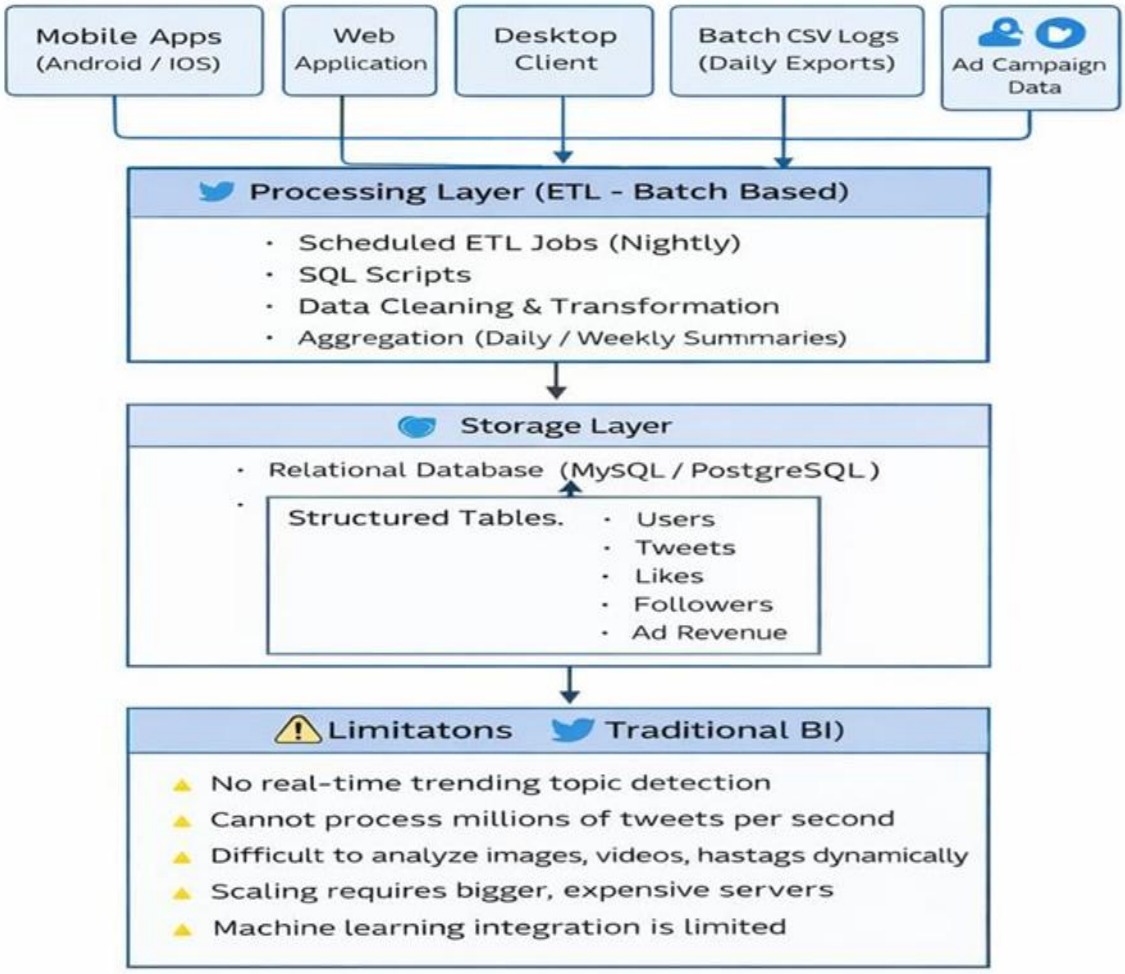

Fig 5.1 – Traditional Data Warehouse

2. Hadoop-Based Big Data Architecture

In the Hadoop-Based Big Data architecture, Twitter data is processed using a distributed and scalable system.

1. Data Ingestion Layer

User activities such as tweets, likes, retweets, comments, and hashtag usage are captured in real time using streaming systems. This allows continuous data flow into the system.

2. Distributed Storage Layer

The incoming data is stored in HDFS (Hadoop Distributed File System), which distributes data across multiple nodes. This ensures fault tolerance and horizontal scalability.

3. Processing Layer

Apache Spark processes the data in parallel across multiple nodes. It performs operations such as:

- Real-time trend detection
- Engagement analysis
- Spam detection
- User behavior analysis
- Spark's in-memory processing enables faster analytics compared to traditional batch systems.

4. Machine Learning Layer

Machine learning models are applied to:

- Predict trending topics
- Recommend relevant tweets
- Identify fake or bot accounts
- Optimize advertisement targeting

5. Visualization Layer

The processed results are displayed using BI dashboards and monitoring systems, enabling quick decision-making.

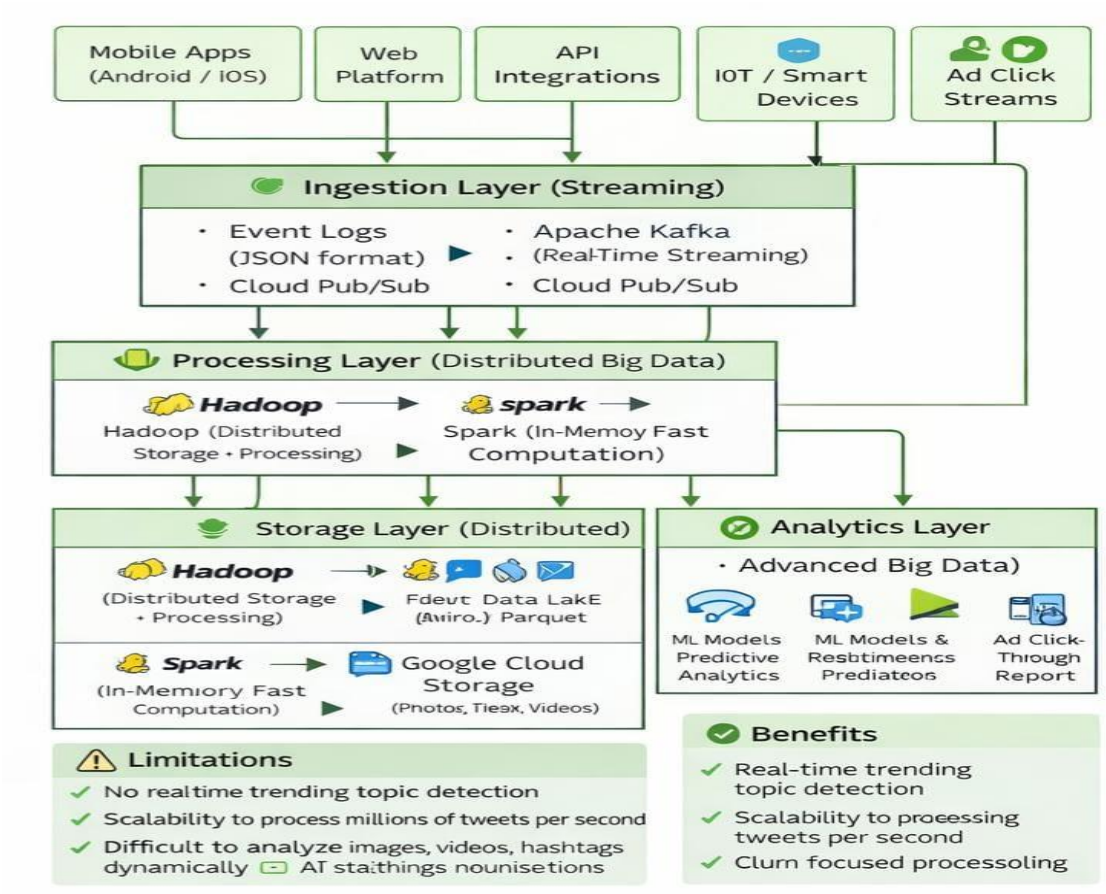

Fig 5.2 – Hadoop based Big Data

6. Results

The results of this project are based on the comparative study between Traditional Data Warehouse Architecture and Hadoop-Based Big Data Architecture in the context of Twitter (X).

1. Comparison Outcome

From the architectural analysis, it is observed that Traditional Business Intelligence systems are suitable only for structured and small-scale data. They rely on centralized storage and batch processing, which limits scalability and real-time analytical capability.

In contrast, the Hadoop-Based Big Data Architecture supports distributed storage and parallel processing. This makes it suitable for handling massive volumes of structured, semi-structured, and unstructured data generated by Twitter every second.

2. Scalability and Performance

The traditional data warehouse model primarily supports vertical scaling, where system capacity is increased by upgrading hardware. This approach increases cost and has physical limitations.

The Big Data architecture demonstrates horizontal scalability, where additional nodes can be added to the cluster to increase storage and processing capacity. This makes the system more flexible, cost-effective, and capable of handling Twitter's rapidly growing data.

3. Real-Time Analytics Capability

Traditional BI systems mainly support batch analytics, where reports are generated at fixed intervals.

The Hadoop-based system enables real-time analytics using distributed processing engines like Apache Spark.

This allows Twitter to:

- Detect trending hashtags instantly
- Monitor live user engagement
- Identify spam or malicious accounts in real time
- Update personalized recommendations dynamically

4. Data Handling Capability

Traditional BI systems primarily handle structured data stored in relational databases.

The Big Data system successfully manages:

- Text-based tweets (unstructured data)
- Images and videos (unstructured data)
- JSON logs and API responses (semi-structured data)
- User interaction records (structured data)
- This demonstrates the flexibility and adaptability of Big Data technologies.

5. Business Impact

The implementation of Big Data architecture provides significant advantages to Twitter:

- Improved user engagement through personalized content
- Faster identification of trending topics
- Enhanced advertisement targeting
- Better decision-making using predictive analytics
- Increased platform scalability for future growth

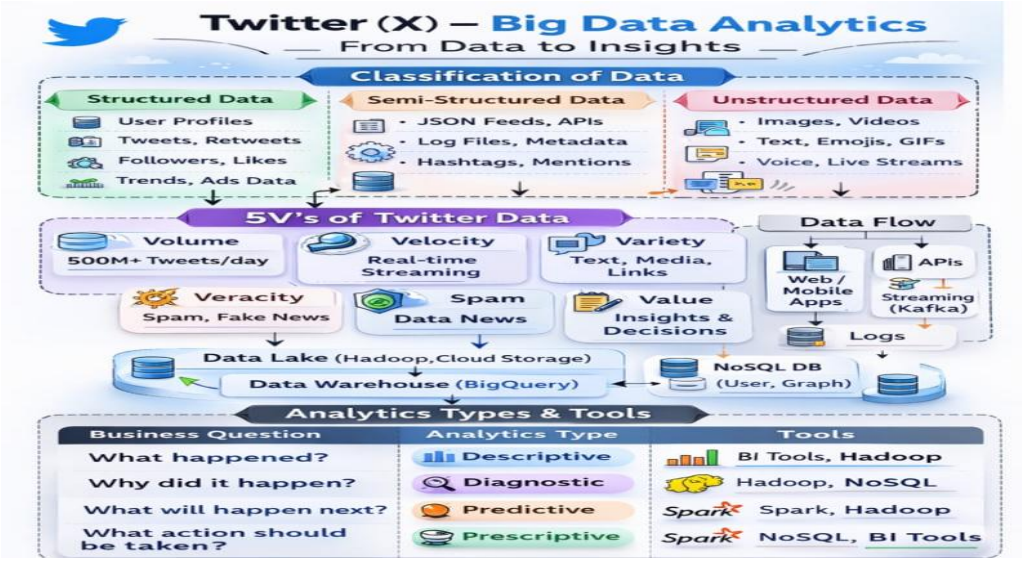

Fig 6.1 – Infographic Poster 1

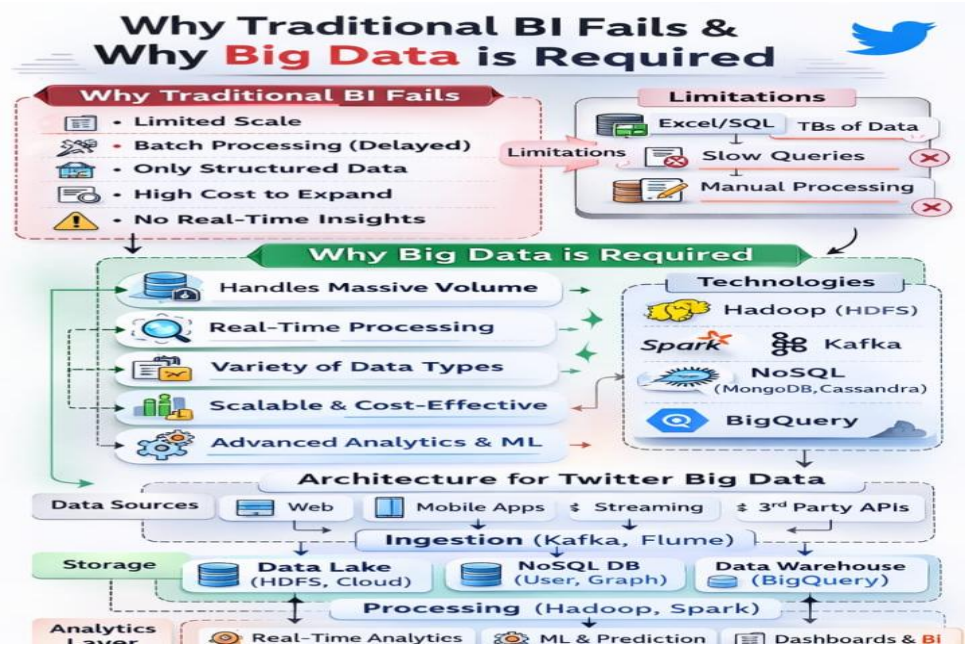

Fig 6.2 – Infographic Poster 2

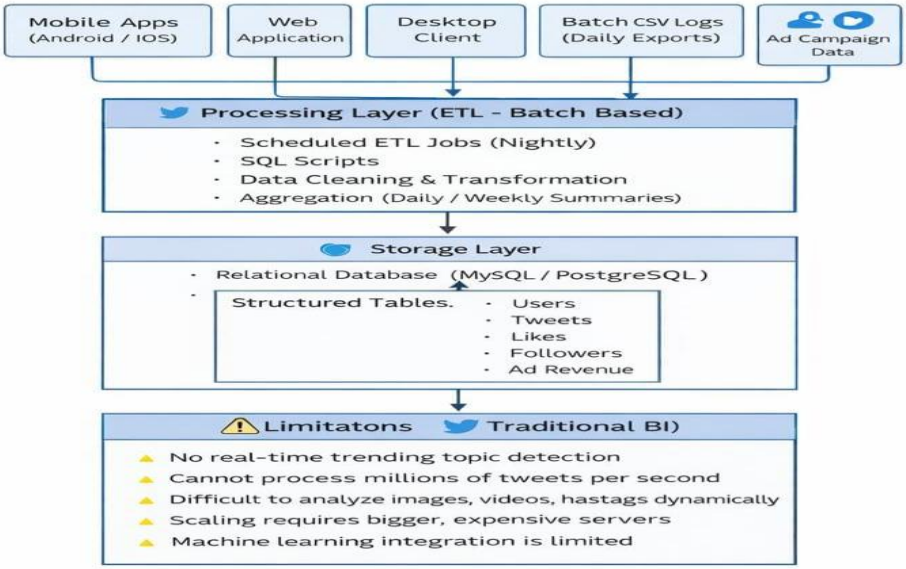

Fig 6. 3 – Traditional Data Warehouse

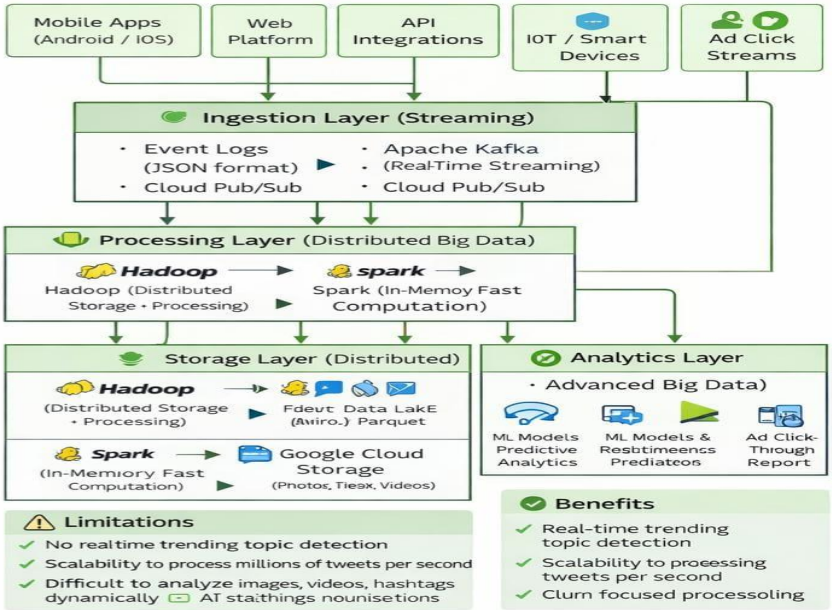

Fig 6.4 – Hadoop Based Data

7. Conclusion

This project successfully demonstrates the importance and necessity of Big Data technologies in modern social media platforms such as Twitter (X). Through the comparative analysis of Traditional Data Warehouse Architecture and Hadoop-Based Big Data Architecture, it is evident that traditional Business Intelligence systems are limited in handling the massive volume, velocity, and variety of data generated in today's digital ecosystem.

While Traditional BI systems are effective for structured data and historical reporting, they lack scalability, flexibility, and real-time processing capabilities required for high-interaction platforms like Twitter. The centralized architecture and batch processing approach make them unsuitable for analysing continuous tweet streams, multimedia content, and dynamic user behavior.

On the other hand, the Hadoop-based Big Data architecture overcomes these limitations by enabling distributed storage through HDFS, parallel data processing using Apache Spark, and flexible data management with NoSQL databases. This architecture supports structured, semi-structured, and unstructured data while ensuring fault tolerance and horizontal scalability. Furthermore, the integration of machine learning models enables predictive and prescriptive analytics, allowing Twitter to detect trending topics, recommend personalized content, identify spam accounts, and optimize advertisement strategies.

The project also highlights the four types of analytics — Descriptive, Diagnostic, Predictive, and Prescriptive — and demonstrates how each type contributes to data-driven decision-making and business growth. Overall, this study establishes that Big Data is not merely a technological upgrade but a strategic requirement for organizations operating at large scale. As digital platforms continue to expand rapidly, the adoption of intelligent and scalable Big Data systems will remain essential for innovation, competitive advantage, and enhanced user experience.